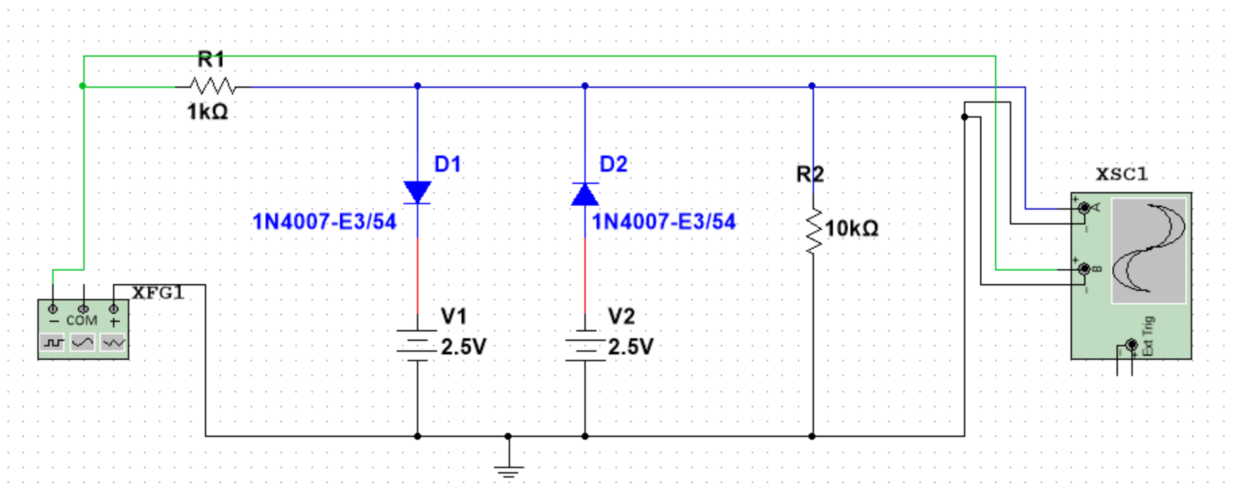
## DOUBLE ENDED CLIPPERS

### CIRCUIT DIAGRAM




# Platform Used

The circuit is designed and simulated using **Multisim**, a widely used electronic circuit simulation software. It allows virtual testing of analog circuits using realistic components such as signal generators, diodes, resistors, DC sources, and oscilloscopes.

# Components Used

1. AC Signal Generator (XFG1)

- Generates the input sinusoidal waveform.
- Provides an alternating voltage that is applied to the clipper circuit.
- The amplitude and frequency can be adjusted as required.

2. Resistor R1 (1 kΩ)

- Connected in series with the input signal.
- Limits the current flowing through the diodes.
- Protects the diodes from excessive current.

3. Diodes D1 and D2 (1N4007)

- Silicon PN junction diodes.
- Used for clipping the positive and negative portions of the input signal.

4. DC Bias Sources V1 and V2 (2.5 V each)

- Provide biasing to the diodes.
- Set the clipping levels for both positive and negative halves of the waveform.

- Make the clipper a **biased double-ended clipper**.

5. Resistor R2 (10 kΩ)

- Acts as the load resistor.
- Output voltage is measured across this resistor.

6. Oscilloscope (XSC1)

- Displays the input and output waveforms.
- Used to observe the clipping action clearly.

## Basic Working of the Double Ended Clipper Circuit

A **double ended clipper circuit** limits both the **positive and negative peaks** of an input signal beyond predefined voltage levels.

### Positive Half Cycle

- When the input voltage exceeds **+ (2.5 V + diode drop)**:
  - Diode **D1 becomes forward biased**.
  - Excess voltage is diverted through D1 and V1 to ground.
  - The output voltage is clipped at the positive level.

### Negative Half Cycle

- When the input voltage goes below **– (2.5 V + diode drop)**:
  - Diode **D2 becomes forward biased**.
  - Excess negative voltage is diverted through D2 and V2.
  - The output voltage is clipped at the negative level.

## Output Observation

- The output waveform appears **flattened at both top and bottom**.
- This confirms that both positive and negative peaks are clipped.
- The clipping levels are determined by the **bias voltages and diode drops**.

## Conclusion

This circuit demonstrates the working of a **biased double ended clipper** using diodes. By applying suitable DC bias voltages, the amplitude of an AC signal can be limited on both sides. Such circuits are commonly used in **signal conditioning, over-voltage protection, and waveform shaping applications.**